

Waseet Logistics Capstone Project

End-to-End Pipeline Orchestration & Historical Analytics Report

1. Architecture & Design Decisions

Relational Schema & Integrity: Built on PostgreSQL 16. The `parcel_scans` table enforces a unique composite constraint on `(parcel_id, scan_type, scanned_at)`, strictly preventing duplicate physical events. An audit table `load_log` tracks run metadata including rows read, loaded, rejected, deduplicated, status, and error traces.

Data Contracts & Schema Drift: `pipeline.py` executes a fail-fast contract check before any transform stage. If incoming CSV headers omit required attributes (such as `weight_kg` on May 19), execution aborts with exit code 1, logging a critical contract violation to prevent warehouse corruption.

Idempotency Strategy: Executed via a staged delete-insert pattern within an atomic transaction: existing records for the scan window are cleared before insertion. Re-running the pipeline on identical input produces identical warehouse state.

Airflow DAG Coordination: `days/waseet_daily.py` implements a `FileSensor` with `mode='reschedule'` (avoiding worker slot starvation), a `BranchPythonOperator` to detect empty/holiday files, and a terminal `EmptyOperator` with `trigger_rule='none_failed_min_one_success'` to prevent clean skips from failing DAG runs.

2. The Four Special Operational Days

Date	Incident Description	Pipeline & Orchestration Response
2026-05-10	Supplier upload failure (file missing).	Sensor polled, reached 30s timeout, triggered failure callback <code>alert_failure</code> . No corrupt data loaded.
2026-05-13	Duplicate delivery file (2,546 rows, 2x volume).	Pipeline extracted payload, identified duplicate scans, purged duplicate events, loaded exactly one copy cleanly.
2026-05-15	Eid Holiday; network closed (0 data rows).	Sensor found file; choose branch routed to <code>skip_day</code> . DAG completed successfully without alerts.
2026-05-19	Schema drift; supplier renamed <code>weight_kg</code> .	Data contract check raised violation prior to transformation. Aborted with exit code 1, logging <code>FAILED</code> to <code>load_log</code> .

3. Quality Suite & 21-Day Reconciliation

Declarative Quality Suite: 10 rules covering Completeness, Validity, Format, and Range. Rules execute against raw incoming CSV files, detecting nulls, invalid scan types, non-positive weights, and malformed ISO timestamps.

Date	File	WH	Diff	Reconciliation Explanation
2026-05-01	1230	1208	22	Standard validation rejects quarantined
2026-05-04	1285	1262	23	3 duplicate scans removed, 20 rejects quarantined
2026-05-10	0	0	0	Supplier delivery failure (file never arrived)
2026-05-13	2546	1248	1298	Concatenated duplicate file; duplicate batch purged
2026-05-15	0	0	0	Eid Holiday (network closed; skipped cleanly)
2026-05-19	1066	0	1066	Run failed: Contract violation (missing <code>weight_kg</code>)
2026-05-21	1222	1200	22	Standard validation rejects quarantined

4. PySpark Historical Analytics & Optimization

Explicit Schema: Enforced via StructType to bypass the expensive double-read penalty of inferSchema=True across 285,000 historical rows.

Broadcast Hash Joins: Small dimensions (hubs.csv with 10 rows and service_levels.csv) were explicitly broadcasted using F.broadcast(). Physical plan inspection confirms BroadcastHashJoin, entirely avoiding costly cross-network cluster shuffle stages.

The Small Files Problem in Parquet:

- Partitioning 18 MB across 334 days yielded 334+ tiny Parquet files (~15-50 KB each).
- *Production Assessment:* Highly suboptimal for HDFS/S3. Micro-files create severe NameNode memory pressure and destroy Parquet's columnar compression ratios.
- *Recommendation:* Retain small datasets unpartitioned or partition monthly (yyyy-MM), targeting 128 MB to 512 MB file chunks.